

LESSON 8.5

EVALUATING NUMERIC EXPRESSIONS – PART 2



LESSON INTRODUCTION

MA.7.NSO.2.1 Solve mathematical problems using multistep order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

MA.7.NSO.2.2 Add, subtract, multiply and divide rational numbers with procedural fluency.

LEARNING TARGETS

- I can evaluate numeric expressions with up to three or four operations.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations.

MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute value.

WORKING TOWARD

MA.8.NSO.1.7 Solve multistep mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

ESSENTIAL QUESTIONS

- How are numeric expressions with multiple grouping symbols evaluated?
- How can exponent laws be used in conjunction with the order of operations?

LESSON NARRATIVE

This lesson is part two of a three-part lesson series. In this lesson, students continue their learning with evaluating numeric expressions. Students will learn how to evaluate expressions with multiple grouping symbols in a given expression as well as apply what they have learned about exponent laws to this process.

COMPONENT SUMMARY

8.5.1 WARM-UP (~5 MIN)

Bubble Wrap estimation activity

8.5.3 EXPLORATION (10-15 MIN)

Collaborative exploration activity on grouping symbols

8.5.5 YOUR TURN! (5-10 MIN)

Independent practice on evaluating expressions

8.5.2 GUIDED INSTRUCTION (10 MIN)

Instruction on evaluating expressions using grouping symbols and rational numbers

8.5.4 COLLABORATION (10 MIN)

Collaborative method comparison activity on the order of operations

8.5.6 WRAP-UP (~5 MIN)

Students reflect on their learning

RESOURCES

GLOSSARY

Key Terms:

- Nested grouping symbols

Additional Terms:

- Order of operations

MATERIALS

- (Optional) Colored markers or highlighters

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse the order of operations by doing all multiplication before division instead of from left to right in order.
- Students may make computation errors when working with decimals without a calculator.

THINGS TO CONSIDER

- Students may have trouble with nested grouping symbols and identifying which numbers/operations are in the inner grouping vs the outer grouping. Consider using colored highlighters or other visual markers to help students see the process unfold.
- Often, there is more than one way to evaluate a numeric expression by applying the order of operations, properties of equality, and laws of exponents. As you work the problems, consider alternate correct ways to evaluate each expression to be prepared for alternate correct suggestions from students.
- Students should not use a calculator for this lesson.
- If time is an issue, then consider assigning 8.5.5 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

In 8.5.2 Guided Instruction, students learn about evaluating expressions when nested grouping symbols are present. To help students understand this, consider demonstrating thinking out loud and prompting students to reflect and respond.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

During 8.5.2 Guided Instruction, students have the opportunity to analyze the mathematical thinking of two students. Discussing the two different methods with their partner helps students construct a justification that reflects the two correct methods. In 8.5.3 Exploration and in 8.5.4 Collaboration, students work with their partner to explore similar numeric expressions where they communicate the order of operations, properties of equality, and laws of exponents.

DIFFERENTIATE INSTRUCTION

8.5.3 Exploration and 8.5.4 Collaboration have students working together in partner pairs. Consider pairing students based on the data from the previous lesson. This can help struggling students learn from their peers while strengthening the other students' understanding of the concept.

8.5.2 Guided Instruction provides instruction on nested grouping symbols. To help with this, have students mark the outer grouping symbols with a highlighter and underline everything between them. Then have them mark the inner grouping symbols with a different colored highlighter and draw a line above the symbols.

8.5.6 Wrap-Up activity in this lesson can provide valuable data on whether or not a student understands the concepts in this lesson.

8.5.1 WARM-UP: ESTIMATING LENGTH

TEACHER MOVES

For this activity, consider using a Think-Pair-Share approach. Allow students to have a minute or two to process the information and answer independently. Then call time and have students turn to their partner and share out their thinking. Call on student pairs to share out what their partner thought and if the pair came to any sort of consensus about their process or answer.

QUESTIONING

For an additional piece of engagement, consider asking students what other products they can think of that would be found in similar structure or packaging as the Bubble Wrap.

ENRICHMENT

How many rolls of the bubble wrap on the right do you think you would need to stretch from one side of the school to the other?



8.5.1 Warm-Up: Estimating Length

The roll of bubble wrap on the left is 1 foot high and 100 feet long.



Complete the following to guess the length of the bubble wrap on the right.

- What is a guess that is too low?
Answers will vary. 100 is too low because the bubble wrap on the left is smaller than the one on the right and is 100 feet long.
- Give a guess that does not make sense and explain how you know it doesn't make sense.
Answers will vary. The bubble wrap is zero feet long. This doesn't make sense because the answer has to be a length.
- What is a guess that is too high?
Answers will vary. The bubble wrap is 1 million feet long.
- What is your best guess of the length of the bubble wrap on the right when unrolled?
Answers will vary. The bubble wrap is double the wrap on the left, so it is 200 feet long.



8.5.2 Guided Instruction: Using the Order of Operations on Expressions with Nested Grouping Symbols

The order of operations can be used to evaluate expressions using grouping symbols, rational numbers, exponents, and absolute values.

1. Two students evaluated the expression $\left(\frac{(3+0.5)}{0.5}\right)^3$. This expression shows nested grouping symbols, or groupings inside of other groupings.

Abby	Devon
$\left(\frac{(3+0.5)}{0.5}\right)^3$	$\left(\frac{(3+0.5)}{0.5}\right)^3$
$\left(\frac{3.5}{0.5}\right)^3$	$\frac{(3+0.5)^3}{0.5^3}$
$(7)^3$	$\frac{3.5^3}{0.5^3}$
343	$\frac{42.875}{0.125}$
	343

How were Abby’s steps and Devon’s steps different from each other?

Answers will vary. Abby solved using order of operations by grouping symbols whereas Devon used the exponent laws before doing any other operation.



When using the order of operations to evaluate an expression with nested grouping symbols, apply the order of operations inside the inner grouping symbols first.

2. Zion says since multiplication is commutative and the order does not matter, he thinks he can evaluate the expression $[0.05 \times (3 \times 2)^4]^2$ by either multiplying 0.05×3 first or by multiplying 3×2 first.

Discuss with your partner if Zion is correct when he stated that both would work.

Answers will vary. The result of the discussion should be that Zion is incorrect. There is an exponent that has to be applied first in this case, so you must multiply 3 by 2 and then raise the result to the 4th power before multiplying by 0.05. Multiplication is commutative; however, what Zion is describing is actually the associative property.

3. Telena says the problem can be worked either by following the order of operations or by using the laws of exponents. Complete the table to show the work for both possibilities to determine if Telena is correct.

Order of Operations	Laws of Exponents
$\left[0.1 \cdot \left(\frac{2}{3} \cdot -3\right)^3\right]^2$	$\left[0.1 \cdot \left(\frac{2}{3} \cdot -3\right)^3\right]^2$
$(0.1 \cdot (-2)^3)^2$ $(0.1 \cdot -8)^2$ $(-0.8)^2$ 0.64	Answers will vary based on law of exponents used. $[0.1^2 \cdot (-2)^6]$ $0.01 \cdot 64$ 0.64

8.5.2 GUIDED INSTRUCTION: USING THE ORDER OF OPERATIONS ON EXPRESSIONS WITH NESTED GROUPING SYMBOLS

TEACHER MOVES

When reviewing the narrative language prior to question 1, consider doing a brief refresher on the vocabulary terms (grouping symbols, rational numbers, exponents, and absolute values). Prompt students to respond to what each term means, what they do, what they look like, and how they are used in the order of operations.

ENRICHMENT

Why do you think that it’s possible for Abby and Devon to go through two different series of steps but still result in the same answer? Do you think that would work in every case? Why or why not?

DIFFERENTIATE INSTRUCTION

Some students may struggle when working with nested grouping symbols. To help with this, have students mark the outer grouping symbols with green highlighter and underline everything between them. Then have them mark the inner grouping symbols with a yellow highlighter and draw a line above the symbols.

TEACHER MOVES

Have students share out their answers to question 3. Look for students who used different exponent laws or a different order to get the answer. Consider charting the different approaches to draw attention to how they all arrived at the same answer.

8.5.3 EXPLORATION: GROUPING SYMBOLS

TEACHER MOVES

Prior to releasing this activity to students, consider reviewing the narrative with the students. Many students will jump straight to the expressions and begin evaluating and miss the key point of them making a guess as to which one they think will have the largest value. You may also want to consider having the class vote and writing the total number of votes per expression in a central area to refer back to when reviewing the answers.



8.5.3 Exploration: Grouping Symbols

- Without working the problems, discuss with your partner which expression in the table you expect to have the largest value.

Answer will vary. Discussion should be around law of exponents and order of operations to result in the largest value.

Expression A	Expression B
$10 - (0.75 \times 12) + 4^2$ $10 - 9 + 4^2$ $10 - 9 + 16$ 17	$(10 - 0.75) \times (12 + 4^2)$ $(9.25) \times (12 + 16)$ 9.25×28 259
Expression C	Expression D
$10 - 0.75 \times 12 + 4^2$ $10 - 0.75 \times 12 + 16$ $10 - 9 + 16$ 17	$10 - 0.75 \times (12 + 4)^2$ $10 - 0.75 \times (16)^2$ $10 - 0.75 \times 256$ $10 - 192$ -182

- Put an asterisk (*) next to the expression you expect will have the largest value. Write your explanation here.

Answer will vary. Explanation will be based on the laws of exponents or order of operations used.

- Evaluate each of the expressions in the table with your partner.

ENRICHMENT

What was the impact of the parentheses on the four expressions in this activity?

- With another set of partners, share if your guess was or was not correct. Discuss how the order of operations affected the value of your guesses. Summarize your discussion.

Answers will vary. The discussion should be about order of operations causing expression D to be the largest value.

- For the two expressions in the table that resulted in the same value, explain why they have the same value.

Where the grouping symbol was placed did not have any effect on the order of operations. Multiplication would have been completed second in both expressions after evaluating the exponent.



8.5.4 Collaboration: Using the Order of Operations to Evaluate Expressions with Nested Grouping Symbols

- The expressions in each row have the same value. Evaluate the expressions in one column while your partner evaluates the expressions in the other column. Compare values and if they are not equivalent, work together to find and correct the error(s).

Partner A	Value of Both Expressions	Partner B
$ (35.5 - 5^2) \div 3 $ $ (35.5 - 25) \div 3 $ $10.5 \div 3$ 3.5	3.5	$ 4.3 \times 5 - 5^2 $ $ 4.3 \times 5 - 25 $ $ 21.5 - 25 $ $ -3.5 $ 3.5
$32 \div [(-2)^2 \times 5 - 4]$ $32 \div [4 \times 5 - 4]$ $32 \div 20 - 4$ $32 \div 16$ 2	2	$[(-2)^3 + 16] \div (-2)^2$ $[-8 + 16] \div 2^2$ $8 \div 4$ 2
$[(18 - 4 \times 5) \times 3]^4$ $[(18 - 20) \times 3]^4$ $[-2 \times 3]^4$ $(-6)^4$ 1296	1296	$[(2.4 - 5.4) \times (-2)^1]^4$ $[-3 \times (-2)^1]^4$ $[-3 \times -2]^4$ 6^4 1296
$11 + (0.3^0 + 3)^2$ $11 + (1 + 3)^2$ $11 + 4^2$ $11 + 16$ 27	27	$[36 \div (2^3 + 4)]^3$ $[36 \div (8 + 4)]^3$ $[36 \div 12]^3$ 3^3 27

8.5.4 COLLABORATION: USING THE ORDER OF OPERATIONS TO EVALUATE EXPRESSIONS WITH NESTED GROUPING SYMBOLS

TEACHER MOVES

When facilitating this activity, look for specific misconceptions that may occur, such as:

- Incorrect order within the grouping symbols.
- Not using the negative symbols when it's being raised to a power.
- Miscalculations when working with decimals by hand.

ENRICHMENT

If students do not have equivalent values, ask the following:

- Did you both follow the order of operations?
- Look at your partner's work. What error in calculation can you find?

8.5.5 YOUR TURN!

DIFFERENTIATE INSTRUCTION

Watch for how students perform on question 1a during this activity. If they are performing poorly, consider pulling the students into small groups to work through one of the other questions and then releasing them to do the rest on their own.



8.5.5 Your Turn!

1. Evaluate each expression without using a calculator.

a. $[(10 - 15) \times 2]^3 + 50$

$$[-5 \times 2]^3 + 50$$

$$(-10)^3 + 50$$

$$-1000 + 50$$

$$-950$$

b. $(2^4 - 3 \times 6)^2$

$$(16 - 3 \times 6)^2$$

$$(16 - 18)^2$$

$$(-2)^2$$

$$4$$

c. $|18 + (-2) \times 14|^5$

$$|18 + (-28)|^5$$

$$|-10|^5$$

$$10^5$$

$$100,000$$

d. $[0.8 \times (1.2 + 2.8)^2]^2$

$$[0.8 \times 4^2]^2$$

$$[0.8 \times 16]^2$$

$$12.8^2$$

$$163.84$$

e. $[(0.8 \times 10)^2 + 4.5] \div 0.5$

$$[8^2 + 4.5] \div 0.5$$

$$[64 + 4.5] \div 0.5$$

$$68.5 \div 0.5$$

$$137$$

f. $(10 \times 0.4)^2 + (28.4 \div (-4))$

$$4^2 + (28.4 \div (-4))$$

$$16 + (-7.1)$$

$$16 + (-7.1)$$

$$8.9$$

**8.5.6 Wrap-Up: Reflection**

1. Complete each statement based on your learning today.

- A barrier to my learning was ...
Answers will vary.
- I will try to overcome this by ...
Answers will vary.
- Today I am still unsure about ...
Answers will vary.
- To fill in this gap I intend to ...
Answers will vary.

8.5.6 WRAP-UP: REFLECTION**DIFFERENTIATE INSTRUCTION**

If students are able to get the correct answer but not identify the mistake, they may not fully comprehend what is occurring incorrectly in someone else's work. If students are not able to locate a mistake in the problem, it will highlight a specific misconception that they have. If students identify the mistakes correctly but the response is not correct, they may have made a calculation error.